

# FIZ1951 FİNAL FORMÜL KAĞIDI

$$\begin{aligned}\varepsilon_o &= 8.85 \times 10^{-12} \frac{F}{m} \\ \mu_o &= 4\pi \times 10^{-7} T \cdot A/m = 1.257 \times 10^{-6} H/m \\ q &= e = 1.6 \times 10^{-19} C \\ h &= 6.63 \times 10^{-34} J \cdot s \quad m_o = 9.11 \times 10^{-31} kg \\ k_B &= 1.38 \times 10^{-23} J/K = 8.617 \times 10^{-5} eV/K \\ 1eV &= 1.6 \times 10^{-19} J \\ \text{Elektron: } m_n &= m_e \\ \text{Boşluk: Hol: Deşik: } m_n &= m_b \\ c &= 3 \times 10^8 \frac{m}{s}\end{aligned}$$

Nötron (n) : Kütle  $m = 1.675 \times 10^{-27} \text{ kg}$ ; Yük  $q = 0$   
 Proton (p) : Kütle  $m = 1.673 \times 10^{-27} \text{ kg}$ ; Yük  $q = +1.602 \times 10^{-19} \text{ C}$   
 Elektron (e) : Kütle  $m = 9.11 \times 10^{-31} \text{ kg}$ ; Yük  $q = -1.602 \times 10^{-19} \text{ C}$

$$\begin{aligned}E &= \frac{hc}{\lambda} \quad \sigma = \frac{ne^2\tau}{m_e^*} \quad \vartheta_t = \sqrt{\frac{3k_B T}{m^*}} \\ E &= \frac{1240 \text{ eV} \cdot \text{nm}}{\lambda(\text{nm})} \quad R = \frac{(n_r - 1)^2 + k^2}{(n_r + 1)^2 + k^2} \\ E &= \frac{1.24 \text{ eV} \cdot \mu\text{m}}{\lambda(\mu\text{m})} \quad n_r = \frac{c}{\vartheta} \\ \alpha(h\nu) &= A(h\nu - E_g)^y \\ I(d) &= I_o \exp(-\alpha d) \\ H &= \frac{B}{\mu} \quad T = (1 - R^2) \exp(-\alpha d) \\ H &= \frac{NI}{l} \quad M = \chi_M H \quad B = \mu_o H + \mu_o M \\ \mu &= \mu_o \mu_r = \mu_o (1 + \chi_M)\end{aligned}$$

$$\begin{aligned}n_i &= p_i \quad \sigma_i = n_i e (\mu_p + \mu_n) \quad n_i = CT^{3/2} e^{-E_g/2k_B T} \\ n_i p_i &= n_i^2 = p_i^2 \quad 1 \geq f_n(E, T) \geq 0 \quad f_n(E, T) = \frac{1}{1 + e^{\frac{E - E_F}{k_B T}}} \quad f_p(E, T) = \frac{1}{1 + e^{\frac{E_F - E}{k_B T}}} \\ N_{CB}(E) &= \frac{1}{2\pi^2 \hbar^3} (2m_e^*)^{3/2} (E - E_C)^{1/2} \quad N_C = 2 \left( \frac{2\pi m_e^* k_B T}{h^2} \right)^{3/2} \quad n_i = N_C \exp\left(-\frac{E_C - E_F}{k_B T}\right) \\ N_{VB}(E) &= \frac{1}{2\pi^2 \hbar^3} (2m_h^*)^{3/2} (E_V - E)^{1/2} \quad N_V = 2 \left( \frac{2\pi m_h^* k_B T}{h^2} \right)^{3/2} \quad p_i = N_V \exp\left(-\frac{E_F - E_V}{k_B T}\right) \\ n_i &= \sqrt{N_C N_V} \exp\left(-\frac{E_g}{2k_B T}\right) \quad \mu_p = \frac{e}{D_p} \quad \mu_n = \frac{e}{D_n} \quad \mu_p = \frac{e}{D_p} \quad \mu_n = \frac{e}{D_n} \\ J_p &= -e D_p \frac{dp}{dx} \quad J_n = e D_n \frac{dn}{dx} \quad E_{Fi} = E_V + \frac{1}{2} E_g + \frac{1}{2} k_B T \ln \frac{N_V}{N_C} \\ D_n &= \mu_n \frac{k_B T}{q} \quad D_p = \mu_p \frac{k_B T}{q} \quad E_{Fi} = E_V + \frac{1}{2} E_g + \frac{3}{4} k_B T \ln \frac{m_h^*}{m_e^*} \\ \sigma_p &= pq\mu_p \quad \sigma_n = nq\mu_n \quad \vartheta_t = \sqrt{\frac{3k_B T}{m^*}} \quad \sigma = \frac{ne^2\tau}{m_e^*} \\ J_{Top} &= \left( en\mu_n E + e D_n \frac{dn}{dx} \right) + \left( ep\mu_p E - e D_p \frac{dp}{dx} \right) \quad N_A \text{ ve } N_D \gg n_i \text{ ve } N_D > N_A \Rightarrow n \text{ tipi} \\ n &= n_i + N_D \quad E_{Fn} = E_C + k_B T \ln \left( \frac{n}{N_C} \right) \quad n = \frac{(N_D - N_A)}{2} + \sqrt{\left( \frac{N_D - N_A}{2} \right)^2 + n_i^2} \\ N_D \gg n_i &\rightarrow n \approx N_D \quad p = \frac{n_i^2}{N_D} \ll N_D \quad n = N_C \exp\left(-\frac{E_C - E_{Fn}}{k_B T}\right) \quad \frac{1}{\mu_n^T} = \frac{1}{\mu_n^L} + \frac{1}{\mu_n^H} \quad \frac{qpV_H}{B_z} = \frac{I_x}{t} \\ p &= \frac{n_i^2}{N_D} \ll N_D \quad \sigma_n \approx N_D e \mu_n \quad \sigma = \frac{1}{ne\mu} \quad l = \frac{e\tau}{m^*} \quad \sigma = 1/\rho \rightarrow S = 1/R \\ p &= \frac{n_i^2}{N_A} \ll N_A \quad N_A \gg p_i \rightarrow p \approx N_A \quad E_{Fp} = E_V - k_B T \ln \left( \frac{p}{N_V} \right) \quad V_{Hall} = R_{Hall} \frac{IB}{t}, \quad R_{Hall} = \frac{1}{nq} \\ p &= \frac{n_i^2}{N_A} \ll N_A \quad \sigma_p \approx N_A e \mu_p \quad p = N_V \exp\left(-\frac{E_{Fp} - E_V}{k_B T}\right) \quad q(\vec{\partial} \times \vec{B}) = q\vec{E} \quad qv_s B = qE_{Hall}\end{aligned}$$

$$\begin{aligned}I_B &= \left( \frac{I_S}{\beta} \right) e^{V_{BB}/V_T} \\ \beta &= \frac{I_C}{I_B}, \quad I_B = \frac{I_C}{\beta} \\ I_B &= \left( \frac{I_S}{\beta} \right) e^{V_{BB}/V_T} \\ I_E &= I_C + I_B \\ I_B &= I_C / \beta \text{ ile, } I_E = \frac{\beta + 1}{\beta} I_C \\ I_C &= \alpha I_E \\ \alpha &= \frac{\beta}{\beta + 1} \\ \beta &= \frac{\alpha}{1 - \alpha} \\ I_D &= I_{DSS} \left( 1 - \frac{V_{GS}}{V_{GS(off)}} \right)^2\end{aligned}$$

$$I = I_o \left( \exp\left(\frac{qV_{D1S}}{nk_B T}\right) - 1 \right) \quad K = \frac{I_{D(on)}}{(V_{GS} - V_{GS(t)})^2}$$

